

CS 736: Medical Image Computing

(Instructor : Suyash P. Awate)

End-Semester Examination

Roll Number: _____

Name: _____

When in doubt, make suitable assumptions, state them clearly, and proceed to solve the problem.
All the best!

Formulae for Reference

Poisson probability mass function: $P(x; \lambda) := \lambda^x \exp(-\lambda)/x!$ with mean = λ , variance = λ , and mode = $\text{floor}(\lambda)$.

Rician probability density function: $P(x; \nu, \sigma) := \frac{x}{\sigma^2} \exp(-\frac{x^2 + \nu^2}{2\sigma^2}) I_0(\frac{x\nu}{\sigma^2}), \forall x \geq 0$

1. [10 points] Suppose you want to use a kernel-based similarity measure to perform k-means clustering of the dataset $\{x_n\}_{n=1}^N$ using distances in the reproducing kernel Hilbert space. Suppose the number of clusters is fixed to some $C > 2$.
 - (5 points) Describe a mathematical formulation for the optimization problem.

- (5 points) Describe an implementable algorithm to solve the optimization problem for the cluster “means” and cluster memberships.

2. [5 points] Considering a Gaussian kernel $k(\cdot, \cdot; \sigma)$ used on a dataset $\{x_n \in \mathbb{R}^9\}_{n=1}^N$, derive the mathematical relationship between (i) the value of the Gaussian-mixture probability density function (PDF) evaluated at $z \in \mathbb{R}^9$ and (ii) the Hilbert-space distance between the points corresponding to the mapped z and the mean of the mapped $\{x_n\}_{n=1}^N$.

3. [25 points] Consider a 2D/planar biological structure whose shape varies across the population. Consider shape represented as a pointset (in pre-shape space) and that shape is an equivalence class of pointsets under similarity transforms. Biology indicates that the shape variability is such that the shapes should be clustered around K different typical shapes.

You want to estimate these K typical shapes $\{\mu_k\}_{k=1}^K$, given a large data sample (comprising N pointsets $\{z_n\}_{n=1}^N$) from the population. However, you don't know which of the given pointsets in the data correspond to which cluster, and so you decide to model a non-binary membership/probability of each pointset to each of the K clusters.

- (10 points) Formulate an optimization problem for the task.

- (15 points) Clearly describe an algorithm to solve this optimization problem.

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4. [20 points] Consider a X-ray computed tomography (CT) imaging scenario. Consider that the following is provided: (i) a matrix of real-valued measurements in the Radon-transform domain and (ii) a matrix modeling all discretized integrals underlying the traversal of the X-ray beams through the scene. Assume that the noise in the data measurements is i.i.d. additive Gaussian.

You want to estimate the image of attenuation coefficients, corresponding to the objects in the scene, from the data.

- (5 points) Mathematically formulate a statistical prior model.

- (5 points) Mathematically formulate a statistical likelihood model.

- (10 points) Mathematically formulate a Bayesian optimization problem, and clearly state an implementable optimization algorithm including initialization (if needed), updates, and termination (if needed).

5. [5 points] Describe the inverse Radon transform algorithmically using the Ram-Lak filter and the Cosine filter.

6. [25 points] Consider you have a statistical model for an grayscale microscopy image X of size 256×256 pixels using a dictionary of A atoms where each atom is a 9×9 pixel grayscale patch. You are told that every 9×9 pixel patch in the uncorrupted image X has a good fit (needn't be perfect) using a small number of atoms in the dictionary model.

Consider an image Y that is a degraded version of image X , such that (i) the imaging system introduces a defocussing blur and (ii) the electronic sensors introduced i.i.d. additive Gaussian noise in the intensity measurements at each pixel in X .

Consider an image Z that is a further degraded version of image Y , such that a central 32×32 pixel region had missing data (i.e., no intensity measurements were possible within that region).

- (5 points) Based on the information provided, propose a prior statistical model $P(X)$.

- (5 points) Propose a likelihood model related to image Y .

- (5 points) Mathematically formulate a Bayesian estimation problem for estimating X given data Y .

- (5 points) Propose a likelihood model related to image Z .

- (5 points) Mathematically formulate a Bayesian estimation problem for estimating X given data Z .

